EXP-2: FORMAL DESIGN OF AN EMBEDDED CONTROLLER USING STATE MACHINE (STATECHART) MODELING – TRAFFIC LIGHT CONTROLLER.

CODE:

```
const int NS_RED = 2;
const int NS_YELLOW = 3;
const int NS_GREEN = 4;

const int EW_RED = 5;
const int EW_YELLOW = 6;
const int EW_GREEN = 7;

enum State {
  NS_GO,
  NS_WAIT,
  EW_GO,
  EW_WAIT
};

State currentState = NS_GO;

void setup() {
  pinMode(NS_RED, OUTPUT);
  pinMode(NS_YELLOW, OUTPUT);
  pinMode(NS_GREEN, OUTPUT);

  pinMode(EW_RED, OUTPUT);
  pinMode(EW_YELLOW, OUTPUT);
```

```
  pinMode(EW_GREEN, OUTPUT);

  Serial.begin(9600);
}

void switchLights(bool nsRed, bool nsYellow, bool nsGreen,
                  bool ewRed, bool ewYellow, bool ewGreen) {
  digitalWrite(NS_RED, nsRed);
  digitalWrite(NS_YELLOW, nsYellow);
  digitalWrite(NS_GREEN, nsGreen);

  digitalWrite(EW_RED, ewRed);
  digitalWrite(EW_YELLOW, ewYellow);
  digitalWrite(EW_GREEN, ewGreen);
}

void loop() {
  switch (currentState) {

    case NS_GO:
      switchLights(LOW, LOW, HIGH, HIGH, LOW, LOW);
      Serial.println("North-South: GREEN | East-West: RED");
      delay(10000);
      currentState = NS_WAIT;
      break;

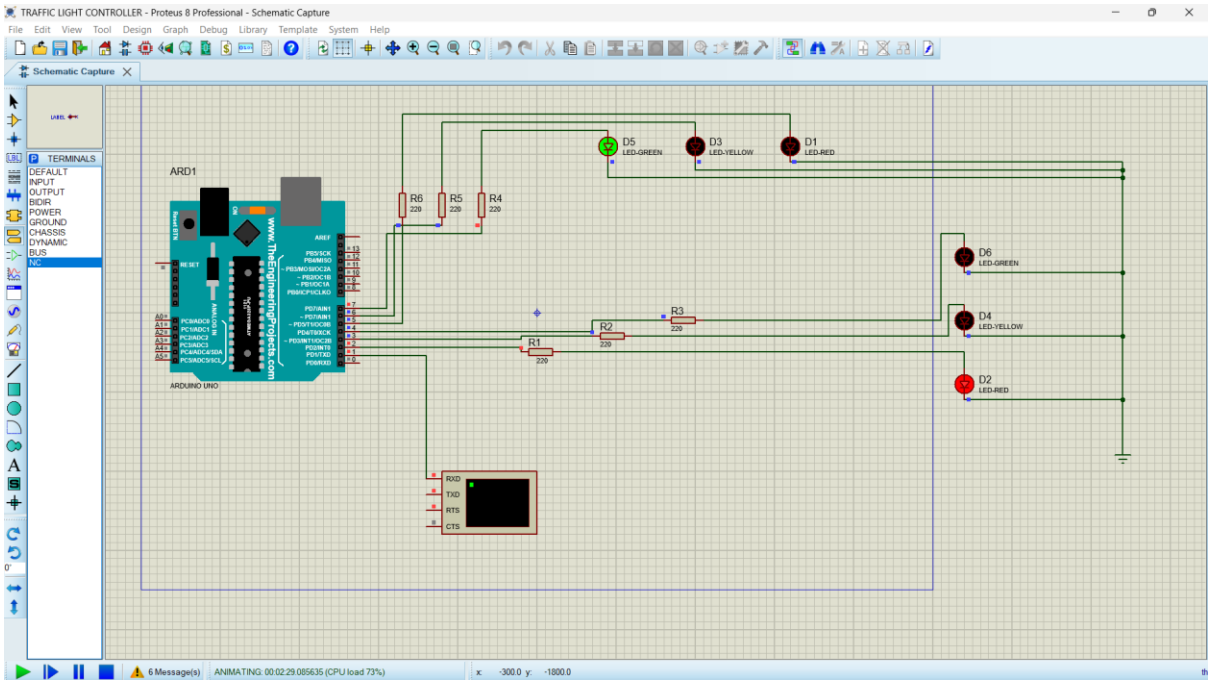
    case NS_WAIT:
      switchLights(LOW, HIGH, LOW, HIGH, LOW, LOW);
      Serial.println("North-South: YELLOW | East-West: RED");
```

```
      delay(3000);
      currentState = EW_GO;
      break;

    case EW_GO:
      switchLights(HIGH, LOW, LOW, LOW, LOW, HIGH);
      Serial.println("North-South: RED | East-West: GREEN");
      delay(10000);
      currentState = EW_WAIT;
      break;

    case EW_WAIT:
      switchLights(HIGH, LOW, LOW, LOW, HIGH, LOW);
      Serial.println("North-South: RED | East-West: YELLOW");
      delay(3000);
      currentState = NS_GO;
      break;
  }
}
```

OUTPUT:


TRAFFIC LIGHT CONTROLLER - Proteus 8 Professional - Schematic Capture
File Edit View Tool Design Graph Debug Library Template System Help
Schematic Capture
TERMINALS
DEFAULT
INPUT
OUTPUT
BIDIR
POWER
GROUND
CHASSIS
DYNAMIC
BUS
NC
ARD1
ARDUINO UNO
www.TheEngineeringProjects.com
R6 220
R5 220
R4 220
R3 220
R2 220
R1 220
D5 LED-GREEN
D3 LED-YELLOW
D1 LED-RED
D6 LED-GREEN
D4 LED-YELLOW
D2 LED-RED
RXD
TXD
RTS
CTS
6 Message(s)
ANIMATING: 00:02:29.085635 (CPU load 73%)
x: -300.0 y: -1800.0